

Lab 05 - Testings

In this lab you will create some unit tests in Java using JUnit.

Simple Math class

Create a class that handle simple math including adding, subtracting, multiplying, and division.

Test of Simple Math class

Create SimpleMathTest class to test the SimpleMath class. It should include:

- Test add positive numbers (ex: $2 + 3 = 5$)
- Test add negative numbers (ex: $-2 + -3 = -5$)
- Test subtract positive numbers (ex: $5 - 3 = 2$)
- Test subtract negative numbers (ex: $-5 - -3 = -2$)
- Test multiply positive numbers (ex: $2 * 3 = 6$)
- Test multiply negative numbers (ex: $-2 * -3 = 6$)
- Test divide positive numbers (ex: $6 / 3 = 2$)
- Test divide negative numbers (ex: $-6 / -3 = 2$)
- Test divide with a/0 (should throw exception)
- Test divide with 0/0 (should throw exception)

Book class

Create a Book class represents book in bookshop.

BookManager class

Create a BookManager class to manage books in bookshop. It should includes:

- List all books (use ArrayList to store books)
- Add new book
- Update existing book by index
- Delete a book by index
- Delete all books

Test of BookManager class

Create class called BookManagerTest that tests all the functions of BookManager class. It should include:

- Test adding new book
- Test updating existing book
- Test deleting a book
- Test deleting all books